

Skills Summary

Distribute, Combine, Factor

Use this reference while completing your worksheet or when studying.

Skill 1 — The distributive property

Rule: $a(b + c) = ab + ac$ — multiply the outside factor by *every* term inside. Picture a rectangle of width $b + c$: the two smaller rectangles ab and ac tile the whole area.

Example: Expand $5(x + 4)$

Step 1. The factor outside the parentheses must multiply every term inside, so distribute it to each term one at a time.

Step 2. $5 \cdot x = 5x$ and $5 \cdot 4 = 20$, so the sum is $5x + 20$.

$$\Rightarrow 5(x + 4) = 5x + 20$$

Try it: Expand $7(x + 2)$

check: $7x + 14$

(!) Watch out: with a negative factor, the sign distributes too — every term inside flips its sign.

Skill 2 — Combining like terms

Rule: like terms have the *same variable part*. Add or subtract their coefficients and keep the variable part unchanged. A constant only combines with another constant.

Example: Simplify $6x + 3 + 2x - 8$

Step 1. Sort the terms into like groups first: the terms with x , and the plain numbers, each combine only with their own kind.

Step 2. x -terms: $6x + 2x = 8x$. Constants: $3 - 8 = -5$.

$$\Rightarrow 6x + 3 + 2x - 8 = 8x - 5$$

Try it: Simplify $9x + 2 - 4x + 6$

check: $5x + 8$

Skill 3 — Factoring out the GCF

Rule: factoring *undoes* distributing. Find the greatest common factor of the terms, then write each term as GCF times something: $ab + ac = a(b + c)$.

Example: Factor $8x + 20$

Step 1. Look for the largest number that divides both coefficients evenly; that number comes outside the parentheses.

Step 2. The GCF of 8 and 20 is 4: $8x = 4 \cdot 2x$ and $20 = 4 \cdot 5$.

$$\Rightarrow 8x + 20 = 4(2x + 5)$$

Try it: Factor $10x + 15$

$(2x + 3)5$:check

(!) Watch out: check your factoring by distributing back — you should land exactly on the expression you started with.